UniGenDet: A Unified Generative-Discriminative Framework for Co-Evolutionary Image Generation and Generated Image Detection

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Research Questions

- How can image generation and synthetic image detection be unified in a single framework to achieve co-evolutionary improvement of both tasks?
- Can detector knowledge improve generation quality while generation understanding enhances detection accuracy and interpretability?
- How to design effective cross-modal attention mechanisms that enable knowledge transfer between generative and discriminative components in a unified model?

Core Idea

Current image generation and detection methods operate independently, leading to an arms race where detectors lag behind evolving generators. This paper proposes the first unified framework where generation and detection co-evolve within a single model. The key insight is bidirectional synergy: the generator's distributional knowledge helps the detector identify subtle artifacts, while the detector's forensic understanding guides the generator to produce more realistic, harder-to-detect images. This is achieved through a Symbiotic Multi-modal Self-Attention mechanism that allows cross-modal information exchange and a two-stage training process with detector-informed generative alignment.

Key Formula

Symbiotic Multi-modal Concatenation

$$h_{concat}^{(l)} = [z_{gen}^{(l)}; h_{det}^{(l)}; h_{text}^{(l)}]$$

Concatenates generation latents z_gen, detection features h_det, and text embeddings h_text at layer l to enable cross-modal attention in the SMSA mechanism

Detector-Informed Generative Alignment Loss

$$L_{DIGA} = E_{x_{GT},\, z_t,\, t}\left[1 - \frac{h_\phi(g(z_t,t)) \cdot f_D(x_{GT})}{\|h_\phi(g(z_t,t))\| \|f_D(x_{GT})\|}\right]$$

Aligns generator features h_φ(g(z_t,t)) with frozen detector features f_D(x_GT) using cosine similarity to guide generator toward producing detector-undetectable outputs

Key Figure

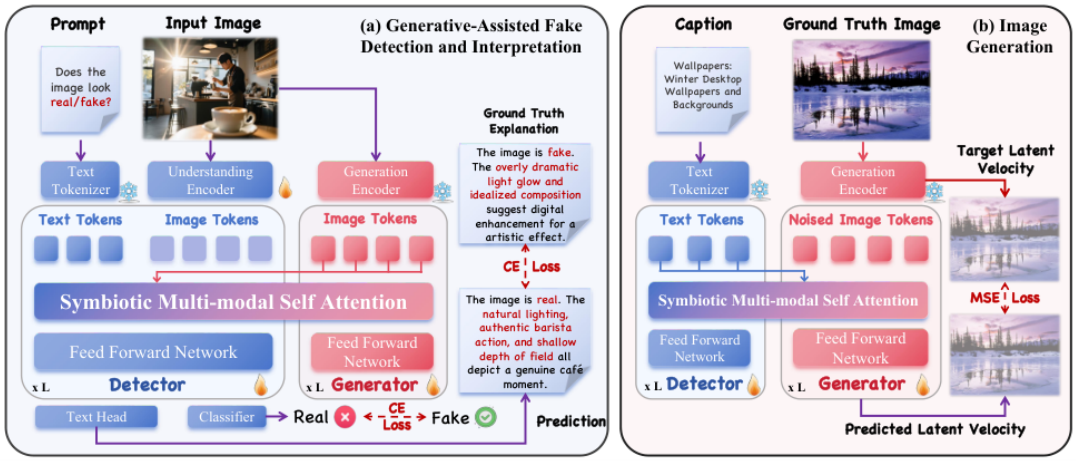

Figure 2. **Overview of the Generation–Detection Unified Fine-tuning (GDUF) pipeline.** (a) *Generative-Assisted Fake Detection and Interpretation:* the **Symbiotic Multi-modal Self-Attention (SMSA)** guides the detector using generator features for authenticity analysis and textual explanation. (b) *Image Generation:* discriminative cues from the detector are injected into the generator for authenticity-aware synthesis. 🔥 denotes trainable modules, and ❄ indicates frozen ones.

Figure 2: Shows the complete GDUF pipeline architecture with SMSA mechanism enabling bidirectional knowledge transfer between generation and detection branches

Limitations

- Requires two-stage training process increasing computational complexity compared to single-task models
- Performance still occasionally fails on highly realistic forgeries or heavily stylized authentic images
- May generate inconsistent textures in highly complex scenes despite forensic guidance
- Limited to specific base architecture (BAGEL) and may not generalize to other unified model architectures